

COMPLETE DOCUMENTATION

RENTipid Unified Autonomous AI Customer Service & Digital Human (v1)

This document serves as the consolidated documentation for the unified AI core.

1. Functional Architecture

- Merges generic customer support (/help) with conversational contextual AI and the Digital Human overlay.
- All actions are processed by the same underlying AiCasePlatform and AiSessionBroker.

2. Technical Architecture

- Framework: Next.js 16
- Database: PostgreSQL (Prisma ORM)
- AI Model: Google GenAI (Gemini 2.5 Pro / Flash)
- Core components: AiSessionBroker, AiCasePlatform, AiToolGateway, AiPolicyEngine, AiCircuitBreaker.

3. Database

- AiSupportCase: Central entity for all support tickets.
- AiConversation: Conversational turn history.
- AiServiceSession: Real-time session data.
- AiCaseEvidence: Attachments and proof for cases.
- AiResolution: Final outcomes and actions.

4. Deterministic Policies

- Refunds, cancellations, payout limits (\$1000/\$500 test thresholds) and damage claims are driven by deterministic, state-machine-backed services (InsuranceReconciliationService, InsuranceCancellationService, etc.). AI does NOT make binding financial decisions.

5. Security & Privacy

- Strict RBAC on tools (AiToolGateway).
- Rate limits on endpoints.
- PII masking applied before sending data to LLM.
- All mutating actions require explicit user confirmation unless pre-approved by policy.

6. Operations & Rollback

- Production readiness: Passing build, successful P12 regression tests.
- Rollback: Supported via schema reversions and feature flags documented in ROLLBACK_VERIFICATION.md.

Closure completed and frozen under commit
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